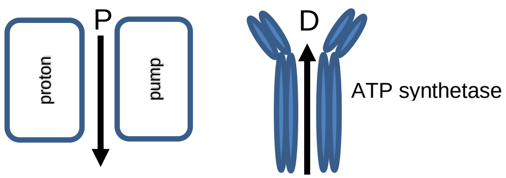
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		**Any five (x1) from** A. In PS II {light / photons} {increase electron energy / excite electrons} (1) B. (higher energy electrons emitted and) accepted by {A_{II} / electron acceptor} / owtte (1) C. {Energy decreases / electrons lose energy} as pass through {ETC / electron carriers} (1) D. (Energy used to) {Pump protons / for ATP production / photophosphorylation} (1) E. PS I {electrons raised to higher energy level / excited} and accepted by {A_I / electron acceptor} (1) F. Use of data (1)	4	1		5		
	(c)	(i)		 **Both correct for 1 mark arrows must be labelled**		1		1		
		(ii)		**Any three (x1) from** • Photolysis releases {H^+ / protons} / photolysis increases concentration of protons} (1) • <u>inside</u> thylakoid <u>{space / lumen}</u> (1) • $NADP \rightarrow NADPH_2$ / NADP {is reduced / decreases the concentration of protons} (1) • <u>in stroma</u> (1)		3		3		
				Question 2 total	**5**	**9**	**2**	**16**	**1**	**2**